1) Consider a case, a folder has multiple files and how would copy it to destination machine path

- To copy a folder containing multiple files within the same Linux machine, the cp command with the -r (recursive) option is used.
- **Example:**

cp -r test_folder /home/hanuman/Desktop/

```
hanuman@hanuman-VMware-Virtual-Platform:~$ cd /home/hanuman/Desktop/
hanuman@hanuman-VMware-Virtual-Platform:~/Desktop$ touch test_folder
hanuman@hanuman-VMware-Virtual-Platform:~/Desktop$ cp test_folder /home/hanuman/
Documents
```

The -r option ensures that all files and subdirectories inside the folder are copied.

To copy a folder from one machine to another over a network using SCP (Secure Copy Protocol), the recursive option -r is also required:

scp -r test_folder user@remote_ip:/destination/path

- **Example:**

scp -r test_folder sri-ragavendra@192.168.23.1:/home/user/Desktop/

```
hanuman@hanuman-VMware-Virtual-Platform:~$ scp The_ESP.webp 'sri-ragavendra\sri ragavendra.s'@192.168.23.1:/C:/Users
/91866/Desktop/
sri-ragavendra\sri ragavendra.s@192.168.23.1's password:
The_ESP.webp                                                  100%   33KB  10.6MB/s   00:00
```

SCP securely transfers files using SSH encryption. It is commonly used to copy directories between Linux systems or between Windows and Linux machines when SSH is enabled.

Now the file The_ESP.webp has been sent from Linux(host device) to Windows(Remote device)

2) Host a FTP and SFTP server and try PUT and GET operations

SCP securely transfers files using SSH encryption. It is commonly used to copy directories between Linux systems or between Windows and Linux machines when SSH is enabled.

The syntax:

ftp <server_ip>

SFTP (Secure File Transfer Protocol) is a secure alternative that operates over SSH. It can be accessed using:

sftp username@server_ip

The PUT command uploads a file to the server:

```
hanuman@hanuman-VMware-Virtual-Platform:~$ sftp 'sri-ragavendra\sri ragavendra.s'@192.168.23.1
sri-ragavendra\sri ragavendra.s@192.168.23.1's password:
Connected to 192.168.23.1.
sftp> put large_files.txt
Uploading large_files.txt to /C:/Users/91866/large_files.txt
large_files.txt                                               100%    0     0.0KB/s   00:00
```

The GET command downloads a file from the server:

```
sftp> get Resume.png
File "/C:/Users/91866/Resume.png" not found.
sftp> get -r /C:/Users/91866/Pictures/Resume.png
Fetching /C:/Users/91866/Pictures/Resume.png to Resume.png
Resume.png                                                    100%   71KB   2.5MB/s   00:00
```

The put and get commands are used inside the SFTP session. Unlike FTP, SFTP encrypts data and credentials, making it more secure.

3. Exploring Network Packets Using Wireshark / TCPdump

Wireshark and tcpdump are packet analysis tools used to monitor network traffic.

Using tcpdump in Linux:

sudo tcpdump -i ens33

```
hanuman@hanuman-VMware-Virtual-Platform:~$ sudo tcpdump -i ens33 port 22 -n
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

To filter SSH traffic:

sudo tcpdump -i ens33 port 22

```
hanuman@hanuman-VMware-Virtual-Platform:~$ sudo tcpdump -i ens33 port 22 -n
[sudo] password for hanuman:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
15:53:58.441087 IP 192.168.23.129.58466 > 192.168.23.1.22: Flags [P.], seq 97984938:97984982, ack 2927072101, win 24
4, options [nop,nop,TS val 3275829424 ecr 5844183], length 44
15:53:58.461041 IP 192.168.23.129.58466 > 192.168.23.1.22: Flags [P.], seq 44:80, ack 1, win 244, options [nop,nop,T
S val 3275829445 ecr 5844183], length 36
15:53:58.461206 IP 192.168.23.1.22 > 192.168.23.129.58466: Flags [.], ack 80, win 250, options [nop,nop,TS val 59043
95 ecr 3275829424], length 0
15:53:58.461263 IP 192.168.23.1.22 > 192.168.23.129.58466: Flags [P.], seq 1:37, ack 80, win 250, options [nop,nop,T
S val 5904396 ecr 3275829424], length 36
15:53:58.461272 IP 192.168.23.129.58466 > 192.168.23.1.22: Flags [.], ack 37, win 244, options [nop,nop,TS val 32758
29445 ecr 5904396], length 0
15:53:58.482731 IP 192.168.23.129.58466 > 192.168.23.1.22: Flags [P.], seq 80:116, ack 37, win 244, options [nop,nop
,TS val 3275829467 ecr 5904396], length 36
15:53:58.483009 IP 192.168.23.1.22 > 192.168.23.129.58466: Flags [P.], seq 37:73, ack 116, win 250, options [nop,nop
,TS val 5904417 ecr 3275829467], length 36
15:53:58.504943 IP 192.168.23.129.58466 > 192.168.23.1.22: Flags [P.], seq 116:152, ack 73, win 244, options [nop,no
p,TS val 3275829489 ecr 5904417], length 36
15:53:58.505253 IP 192.168.23.1.22 > 192.168.23.129.58466: Flags [P.], seq 73:109, ack 152, win 250, options [nop,no
p,TS val 5904440 ecr 3275829489], length 36
```

Wireshark provides a graphical interface where filters such as:

ip.addr == 192.168.1.1
tcp.port == 22
icmp

37	32.527421	fe80::243c:12c4:d20…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
38	32.527883	192.168.88.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
39	32.606929	fe80::243c:12c4:d20…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
40	32.624792	192.168.88.1	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
41	32.640564	fe80::243c:12c4:d20…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
42	32.640756	192.168.88.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
43	32.641075	fe80::243c:12c4:d20…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
44	32.641296	192.168.88.1	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
45	32.642625	fe80::243c:12c4:d20…	ff02::1:3	LLMNR	94	Standard query 0x2cf0 ANY Sri-Ragavendra
46	32.642724	192.168.88.1	224.0.0.252	LLMNR	74	Standard query 0x2cf0 ANY Sri-Ragavendra
47	32.671113	192.168.88.1	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
48	32.671240	fe80::243c:12c4:d20…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
49	33.055257	fe80::243c:12c4:d20…	ff02::1:3	LLMNR	94	Standard query 0x2cf0 ANY Sri-Ragavendra
50	33.055319	192.168.88.1	224.0.0.252	LLMNR	74	Standard query 0x2cf0 ANY Sri-Ragavendra

4. Understanding Linux Networking Commands (ping, arp, ifconfig)

- The ping command checks connectivity between devices:
  ping 192.168.23.1

```
hanuman@hanuman-VMware-Virtual-Platform:~$ ping 192.168.23.1
PING 192.168.23.1 (192.168.23.1) 56(84) bytes of data.
64 bytes from 192.168.23.1: icmp_seq=1 ttl=128 time=0.359 ms
64 bytes from 192.168.23.1: icmp_seq=2 ttl=128 time=0.250 ms
64 bytes from 192.168.23.1: icmp_seq=3 ttl=128 time=0.227 ms
64 bytes from 192.168.23.1: icmp_seq=4 ttl=128 time=0.269 ms
64 bytes from 192.168.23.1: icmp_seq=5 ttl=128 time=0.280 ms
64 bytes from 192.168.23.1: icmp_seq=6 ttl=128 time=0.274 ms
64 bytes from 192.168.23.1: icmp_seq=7 ttl=128 time=0.232 ms
64 bytes from 192.168.23.1: icmp_seq=8 ttl=128 time=0.202 ms
64 bytes from 192.168.23.1: icmp_seq=9 ttl=128 time=0.227 ms
64 bytes from 192.168.23.1: icmp_seq=10 ttl=128 time=0.278 ms
64 bytes from 192.168.23.1: icmp_seq=11 ttl=128 time=0.231 ms
64 bytes from 192.168.23.1: icmp_seq=12 ttl=128 time=0.302 ms
```

It uses ICMP packets to verify reachability

The ping 192.168.23.1 demonstrates successful Layer 3 connectivity between your Linux VM and the Windows host. Each reply line shows an ICMP Echo Reply with low round-trip times (~0.2–0.3 ms), indicating both systems are on the same local virtual network. The TTL value of 128 confirms the responding system is Windows, since Windows uses 128 as its default TTL. Because the latency is extremely low and no hops are involved, this communication is occurring directly within the VMware NAT network, not through an external router.

- The arp -a command displays the ARP table, which maps IP addresses to MAC addresses.

```
hanuman@hanuman-VMware-Virtual-Platform:~$ arp -a
_gateway (192.168.23.2) at 00:50:56:fd:ca:40 [ether] on ens33
? (192.168.23.254) at 00:50:56:f8:c0:a4 [ether] on ens33
? (192.168.23.1) at 00:50:56:c0:00:08 [ether] on ens33
```

The arp -a shows the ARP table, which maps IP addresses to MAC addresses at Layer 2. When you pinged 192.168.23.1, your Linux VM resolved its MAC address and stored it in the ARP cache. The entry 192.168.23.1 at 00:50:56:c0:00:08 corresponds to the Windows VMnet8 adapter. The entry _gateway (192.168.23.2) represents the VMware NAT gateway. The MAC prefix **00:50:56** belongs to

VMware, confirming these are virtual devices. Together, these three outputs verify that your interface configuration, IP communication, and MAC resolution are all functioning correctly within the virtual network.

The ifconfig or ip addr command shows:

- IP address
- Subnet mask
- Broadcast address
- MAC address
- MTU
- Interface status

These commands are essential for network troubleshooting.

```
hanuman@hanuman-VMware-Virtual-Platform:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.23.129  netmask 255.255.255.0  broadcast 192.168.23.255
        inet6 fe80::20c:29ff:fe30:d521  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:30:d5:21  txqueuelen 1000  (Ethernet)
        RX packets 7650  bytes 4363488 (4.3 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 5892  bytes 629407 (629.4 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 250  bytes 64714 (64.7 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 250  bytes 64714 (64.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

The ipconfig shows the configuration of your Linux VM's network interfaces. The active interface is ens33, which has the IPv4 address 192.168.23.129 with a subnet mask of 255.255.255.0 (/24), meaning it belongs to the network 192.168.23.0–192.168.23.255. The broadcast address is 192.168.23.255, and the MTU is 1500 bytes, which is the standard Ethernet frame size. The MAC address (00:0c:29:30:d5:21) identifies your virtual network card. The flags <UP,BROADCAST,RUNNING,MULTICAST> confirm the interface is enabled and functioning. The RX/TX statistics show packets being transmitted and received without errors, indicating a healthy network link. The lo interface is the loopback (127.0.0.1), used for internal communication within the same machine.

5. What Happens When Duplicate IP Addresses Are Configured

When two devices are assigned the same IP address in the same network, it causes an IP conflict. Both devices respond to ARP requests for that IP, leading to unstable connectivity. Network communication becomes intermittent, and users may experience disconnections. Modern systems detect duplicate IP addresses and display warnings. DHCP reduces such conflicts by centrally managing IP allocation.

The ipconfig shows the configuration of your Linux VM's network interfaces. The active interface is ens33, which has the IPv4 address 192.168.23.129 with a subnet mask of 255.255.255.0 (/24), meaning it belongs to the network 192.168.23.0–192.168.23.255. The broadcast address is 192.168.23.255, and the MTU is 1500 bytes, which is the standard Ethernet frame size. The MAC address (00:0c:29:30:d5:21) identifies your virtual network card. The flags <UP,BROADCAST,RUNNING,MULTICAST> confirm the interface is enabled and functioning. The RX/TX statistics show packets being transmitted and received without errors, indicating a healthy network link. The lo interface is the loopback (127.0.0.1), used for internal communication within the same machine.

6. Accessing Remote Systems Using VNC, AnyDesk, TeamViewer, and RDP

Remote systems can be accessed using:

- **VNC** (port 5900)
- **RDP** (port 3389)
- **AnyDesk / TeamViewer** (cloud-based remote access)

RDP can be accessed in Windows using:

mstsc

VNC requires a VNC server on the remote machine. AnyDesk and TeamViewer provide remote access via unique IDs and do not require manual port configuration.
These tools allow remote desktop control over local or internet networks.

7. Checking Default Gateway Reachability

The default gateway can be identified using:

Linux:

ip route

```
hanuman@hanuman-VMware-Virtual-Platform:~$ ping 192.168.23.1
PING 192.168.23.1 (192.168.23.1) 56(84) bytes of data.
64 bytes from 192.168.23.1: icmp_seq=1 ttl=128 time=0.287 ms
64 bytes from 192.168.23.1: icmp_seq=2 ttl=128 time=0.226 ms
64 bytes from 192.168.23.1: icmp_seq=3 ttl=128 time=0.224 ms
64 bytes from 192.168.23.1: icmp_seq=4 ttl=128 time=0.228 ms
64 bytes from 192.168.23.1: icmp_seq=5 ttl=128 time=0.257 ms
64 bytes from 192.168.23.1: icmp_seq=6 ttl=128 time=0.235 ms
```

Windows:

ipconfig

```
C:\Users\91866>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 8:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::243c:12c4:d200:78ee%22
   IPv4 Address. . . . . . . . . . . : 192.168.88.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::4e72:6700:c160:7f76%11
   IPv4 Address. . . . . . . . . . . : 192.168.23.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : VITCHOSDNS
   Link-local IPv6 Address . . . . . : fe80::980b:ecd5:8af3:6d2c%8
   IPv4 Address. . . . . . . . . . . : 172.20.24.199
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 172.20.24.1
```

To check if it is reachable:

ping <gateway_ip>

If the gateway responds, it confirms that routing to external networks is possible. The default gateway acts as the router that forwards traffic outside the local subnet.

8. Checking Interface Speed, MTU, and Network Parameters

In Linux:

ip link show

```
hanuman@hanuman-VMware-Virtual-Platform:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mod
e DEFAULT group default qlen 1000
    link/ether 00:0c:29:30:d5:21 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
```

This displays the MTU value (usually 1500 bytes).

To check interface speed:

sudo ethtool ens33

```
hanuman@hanuman-VMware-Virtual-Platform:~$ sudo ethtool ens33
[sudo] password for hanuman:
Settings for ens33:
        Supported ports: [ TP ]
        Supported link modes:   10baseT/Half 10baseT/Full
                                100baseT/Half 100baseT/Full
                                1000baseT/Full
        Supported pause frame use: No
        Supports auto-negotiation: Yes
        Supported FEC modes: Not reported
        Advertised link modes:  10baseT/Half 10baseT/Full
                                100baseT/Half 100baseT/Full
                                1000baseT/Full
        Advertised pause frame use: No
        Advertised auto-negotiation: Yes
        Advertised FEC modes: Not reported
        Speed: 1000Mb/s
        Duplex: Full
        Auto-negotiation: on
        Port: Twisted Pair
        PHYAD: 0
        Transceiver: internal
        MDI-X: off (auto)
        Supports Wake-on: d
        Wake-on: d
        Current message level: 0x00000007 (7)
                               drv probe link
        Link detected: yes
```

This shows:

- Speed (e.g., 1000Mb/s)
- Duplex mode
- Link detected status

These parameters help determine network performance and diagnose issues such as fragmentation or slow connectivity.

9. Logging into Router Web Interface and Checking Connected Devices

The router IP is identified using:

ipconfig (Windows)
ip route (Linux)

The default gateway IP (e.g., 192.168.1.1 or 172.20.24.1) is entered in a browser:

http://<gateway_ip>

After logging in with router credentials, the connected devices list can be viewed under sections like:

- DHCP Client List
- Connected Devices
- LAN Clients

In managed networks (e.g., hostel WiFi), access to the router interface may be restricted.

10. How DHCP Assigns IP Addresses

DHCP automatically assigns IP addresses using the DORA process:

1. Discover – Client broadcasts request.
2. Offer – Server proposes an IP address.
3. Request – Client accepts the offered IP.
4. Acknowledge – Server confirms assignment.

The DHCP server also provides subnet mask, default gateway, DNS server, and lease time. This prevents IP conflicts and simplifies network configuration.

DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network configuration parameters to devices when they connect to a network. It works using a four-step process called DORA: first, the client sends a Discover message as a broadcast to find a DHCP server; the server responds with an Offer containing an available IP address along with subnet mask, default gateway, DNS server, and lease time; the client then sends a Request message accepting the offered IP; finally, the server sends an Acknowledge (ACK) confirming the assignment. The IP address is given for a specific lease period and must be renewed before expiration. In your hostel network, the router (172.20.24.1) acts as the DHCP server and automatically assigned your system the IP address 172.20.24.199 along with its subnet mask and gateway, eliminating the need for manual configuration and preventing IP conflicts.

11. Connecting to a Remote Machine Using SSH and Telnet

SSH (Secure Shell) is used for secure remote login:

ssh username@remote_ip

The connection from Linux to Windows:

```
hanuman@hanuman-VMware-Virtual-Platform:~$ ssh 'sri-ragavendra\sri ragavendra.s'
@172.20.24.199
sri-ragavendra\sri ragavendra.s@172.20.24.199's password:

Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

sri ragavendra.s@SRI-RAGAVENDRA C:\Users\91866>
```

The connection from Windows to Linux:

```
C:\Users\91866>ssh hanuman@192.168.23.129
hanuman@192.168.23.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

139 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

10 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Mon Mar  2 14:43:29 2026 from 192.168.23.1
hanuman@hanuman-VMware-Virtual-Platform:~$
```

SSH uses encryption (port 22) and is widely used for secure administration.

Telnet uses:

telnet remote_ip

Telnet operates on port 23 and allows remote terminal access. However, it is insecure because it transmits data in plain text. Modern systems disable Telnet server by default, and SSH is preferred for secure communication.